

Signals and Systems

Sessional-II Exam

(EE2008)

Total Time (Hrs): 1
Total Marks: 20
Total Questions: 2

Date: April 8, 2026

Course Instructor(s)

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Roll No

4B-1

Section

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Student Signature

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1. Attempt all questions. *Hints* are given in italics.
2. Show all the steps with the help of labelled diagrams and mathematical equations.

CLO # 03: Apply Fourier series and Fourier transform representation techniques to signals encountered in engineering domain.

- Q1:** A full-wave rectified sine wave is depicted in the Figure 1 below. Calculate the exponential Fourier series and draw the corresponding Fourier spectrum D_n vs ω . [10]

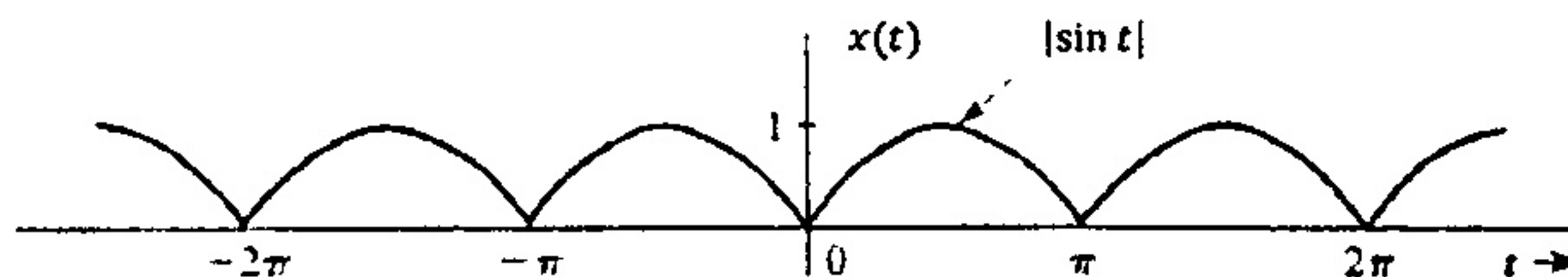


Figure 1. Full-wave Rectified Sine Wave

Hint: $\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$ OR $\sin x = \frac{(e^{jx} - e^{-jx})}{2j}$

CLO # 03: Apply Fourier series and Fourier transform representation techniques to signals encountered in engineering domain.

- Q2:** Determine the Fourier transform of the modulated signal $x(t) \cos 20t$ in which $x(t)$ is a modulating signal of unit height and pulse width of '8' centered about the origin and $\cos 20t$ is a carrier signal. Illustrate with the help of diagrams in both time and frequency domains. [10]

Hint: $x(t) = \text{rect}(t/\tau) \Leftrightarrow \tau \text{sinc}(\omega\tau/2)$ and frequency-shifting property $x(t)e^{j\omega_0 t} \Leftrightarrow X(\omega - \omega_0)$ can be used.